

LLMs struggle to reconstruct masked prompts in complex, domain-specific tasks but perform better in simpler common tasks, highlighting the varying levels of internal knowledge within LLMs across different domains.

Besides, the counterintuitive results on the NoRa-Commonsense task may be attributed to task-specific characteristics. Observation of reconstructed masked prompts in this task indicates that LLMs tend to bypass reconstruction instructions in noisy settings, directly providing final answers. This accidental removal of all rationales, including noisy ones, effectively acts as a noise filter. Combined with the LLM’s natural strength in the commonsens task, this unplanned filtering likely explains the small improvement in accuracy under noisy conditions.

The original clean rationale

Carrie is Hugh’s sister’s husband’s daughter, so the relations path is sister, husband, daughter. For husband’s daughter, we have husband’s daughter is daughter. So the relations path are reduced to sister, daughter. For sister’s daughter, we have sister’s daughter is niece. So the relations path are reduced to niece. Therefore, Answer: niece.

The rephrased rationale by SD

To determine Carrie’s relationship to Hugh, we can break down the given information step by step: 1. Carrie is Hugh’s sister’s husband’s daughter. 2. Hugh’s sister is Carrie’s sibling. 3. Hugh’s sister’s husband is Carrie’s brother-in-law. 4. Carrie is the daughter of Hugh’s sister’s husband. Therefore, the answer is: Carrie is Hugh’s niece.

The original noisy rationale

Carrie is Hugh’s sister’s husband’s daughter, so the relations path is sister, husband, daughter. For husband’s daughter, we have husband’s daughter is daughter. We have daughter’s grandmother is niece. So the relations path are reduced to sister, daughter. For sister’s daughter, we have sister’s daughter is niece. We have daughter’s niece is niece. So the relations path are reduced to niece. Therefore, Answer: niece.

The rephrased rationale by SD

niece.

Table 36: Rephrasing performance of SD under different noisy settings.

The original clean rationale

Francisco is Lynn’s son’s father’s brother, so the relations path is son, father, brother. For father’s brother, we have father’s brother is uncle. So the relations path are reduced to son, uncle. For son’s uncle, we have son’s uncle is brother. So the relations path are reduced to brother. Therefore, Answer: brother.

The rephrased rationale by SD

To determine Francisco’s relationship with Lynn, we start with the information provided: Francisco is Lynn’s son’s father’s brother. Step 1: Lynn’s son’s father is Lynn’s husband. Step 2: Francisco is Lynn’s husband’s brother. Based on this information, we can conclude that Francisco is Lynn’s brother-in-law. Answer: brother-in-law

The original noisy rationale

Francisco is Lynn’s son’s father’s brother, so the relations path is son, father, brother. For father’s brother, we have father’s brother is uncle. So the relations path are reduced to son, uncle. For son’s uncle, we have son’s uncle is brother. We have uncle’s niece is nephew. So the relations path are reduced to brother. We have brother’s nephew is mother. Therefore, Answer: brother.

The rephrased rationale by SD

Answer: brother.

Table 37: Rephrasing performance of SD under different noisy settings.

F.4 The Difficulty of Irrelevant Noise Semantics

We conduct extra experiments with different semantic difficulties. Specifically, we set up three levels of semantic difficulty:

- **Level-1: Topic-irrelevant.** Noisy thought has little relevance to the task. Here, noisy thought involves concepts that are clearly irrelevant to the knowledge required for the task. For example, (1) in physics, the number one is the charge of a single unit of positive or negative electricity; (2) in chemistry, two is the atomic number of helium, the second most abundant element in the universe. These two are irrelevant thoughts.
- **Level-2: Topic-relevant and task-irrelevant.** The noisy thought has some apparent relevance to the task but does not help solve the problem. Noisy thought may involve some terminology or concepts that are relevant to the task but do not provide useful information.

For example, (1) the modulo operation (mod) finds the remainder after the division of one number by another. (2) In base-9, the leading digit 1 indicates that the number is greater than or equal to $1 * 9^2 = 81$ but less than $2 * 9^2 = 162$. (3) In base-9, the number 9 is represented as '10' since it is the base value itself. (4) The number 11 in base-9 is equivalent to the decimal value ninety-nine, which is one less than the square of the base value.

- **Level-3: Topic-relevant, task-relevant, but not helpful.** Here, the noisy thought has some relevance to the task, but is not necessary information to solve the problem. The noisy thought may involve concepts, principles, or methods that are relevant to the task, but not directly applicable to the problem at hand.

For example, (1) in a base-9 system, the place values are units, nines, eighty-ones, and so on, analogous to units, tens, hundreds, etc., in base-10; (2) the concept of carrying is essential when performing addition in positional numeral systems like base-9 and base-10. The modulo operation is a crucial step in the process of adding numbers in base-9, as it determines the digit and carry values; (3) the modulo operation is a crucial step in the process of adding numbers in base-9, as it determines the digit and carry values.

Based on the above definition and prompt, we generate the data with the following four steps.

- **Step-1: Initial generation and human evaluation.** We used this prompt to generate an initial set of irrelevant statements for each level. These were then manually reviewed and filtered to ensure they accurately represented the intended level of irrelevance. We selected high-quality examples for each level.
- **Step-2: Scaled generation.** Using these high-quality examples as ICL demonstrations, we prompted GPT-4 to generate a larger set of irrelevant statements for each level.
- **Step-3: Validation through similarity analysis.** To confirm that our generated statements indeed represented different levels of irrelevance, we conducted a cosine similarity analysis. This analysis quantitatively demonstrated the semantic differences between levels, as shown in our previous response.
- **Step-4: Dataset construction.** we integrated these generated irrelevant statements into our dataset. We inserted them into relevant demonstrations at appropriate positions, following a probability distribution that corresponds to the intended difficulty level of the task.

The experiment results with the above irrelevant thoughts are elaborated as follows: Here, the setup is kept the same as the main experiments to be consistent. Therein, we test the GPT-3.5 model with 3-shot irrelevant rationales where the noise ratio is 0.8.

For a reasonable quantification of relevance, we choose the metric of cosine similarity with the [embeddings API](#) provided by OpenAI with model: text-embedding-ada-002. We calculate the cosine similarity between all noise-inserted sample sentences and the original clean inference sentences and then take their average value.

Dataset	Semantic level	Method	Acc(%)
Math Base-9	Level-1 (Cosine Similarity = 0.75)	base model CD-CoT	26.6 58.3
	Level-2 (Cosine Similarity = 0.87)	base model CD-CoT	30.6 60.0
	Level-3 (Cosine Similarity = 0.88)	base model CD-CoT	33.0 59.7
Symbolic Equal	Level-1 (Cosine Similarity = 0.73)	base model CD-CoT	23.0 44.0
	Level-2 (Cosine Similarity = 0.79)	base model CD-CoT	26.9 46.0
	Level-3 (Cosine Similarity = 0.82)	base model CD-CoT	27.7 45.0

Table 38: Performance of GPT-3.5-turbo under different noise semantics.

For comparison, the accuracy when prompting with clean rationales is 46.4% for NoRa-Math Base-9 and 32.7% for NoRa-Symbolic Equal. Specifically, we have the following observations:

Observation F.5 (CD-CoT method is still effective under different semantic levels of noise). Notably, we further verify that the proposed CD-CoT method is still effective under this imperfect scenario with different semantic levels of noise. Specifically, CD-CoT can achieve 58.3%, 60.0%, and 59.7% accuracy on the three semantic levels of noise for the NoRa-Math Base-9 tasks and 44.0%, 46.0%, and 45.0% accuracy on the NoRa-Symbolic Equal tasks. Compared to the base model, CD-CoT shows a significant improvement of 29.4% and 19.1% in accuracy on semantic level-2 reasoning for the NoRa-Math Base-9 and NoRa-Symbolic Equal tasks, respectively. Similarly, CD-CoT demonstrates a 26.7% and 17.3% increase in accuracy on semantic level-3 reasoning for the NoRa-Math Base-9 and NoRa-Symbolic Equal tasks.

Observation F.6 (CD-CoT maintains a robust and stable performance across different semantic levels of noise.). As the semantic level of noise increases, the base model exhibits a certain level of robustness to the irrelevant noise but still suffers a considerable performance drop (15.8% and 5.8% decrease for semantic level-2, and 13.4% and 5.0% decrease for semantic level-3 on the NoRa-Math Base-9 and NoRa-Symbolic Equal tasks, respectively). In contrast, CD-CoT maintains a robust and stable performance across different semantic levels of noise.

F.5 The Number of Noisy Thoughts

As introduced in the main text, we use noise ratios of 0.3, 0.5, and 0.8 to control the number of noisy thoughts. The statistics of the NoRa dataset are shown in Tab. 2, showing the exact number of clean/noisy thoughts in each task. NoRa-Math entails 8 thoughts, NoRa-Symbolic varies from 2 to 12, and NoRa-Commonsense requires 5 thoughts.

To figure out the exact number of noisy thoughts for the model to be robust, we conduct further experiments to cover a wider range of noisy thoughts. Specifically,

- We compare the model’s performance on NoRa-Math Base-9 and NoRa-Symbolic Equal tasks using clean and noisy rationales containing varying numbers of irrelevant or inaccurate thoughts.
- We gradually increase the number of noisy thoughts from 1 to 6 to investigate the impact of the noise level on the model’s accuracy.
- The total number of thoughts per shot, including both clean and noisy thoughts, is shown in parentheses for the NoRa-Math Base-9 task.
- For the NoRa-Symbolic Equal task, the number of thoughts in the clean demonstrations varies across different examples.
- To ensure a consistent comparison, we select a subset of examples with a similar number of thoughts (ranging from 8 to 12) for both clean and noisy rationale experiments.
- We report the average number of thoughts (10.2) for the clean rationales in this subset.
- Consequently, the total number of thoughts per shot for noisy rationales in the NoRa-Symbolic Equal task is also calculated based on this average.

According to the experimental results in the Tab. 39, we can observe that:

Observation F.7 (The impact of noisy thoughts varies across different noise types). For the NoRa-Math Base-9 task, the model’s accuracy remains relatively stable when only one irrelevant noisy thought is introduced. However, the accuracy drops significantly as the number of irrelevant noisy thoughts increases. On the other hand, for the NoRa-Symbolic Equal task, even a single irrelevant or inaccurate noisy thought can lead to a noticeable decrease in accuracy.

Observation F.8 (Inaccurate noisy thoughts have a more severe impact on the model’s performance compared to irrelevant noisy thoughts). In the NoRa-Math Base-9 task, the accuracy drops drastically from 46.4% to 34.8% with just one inaccurate, noisy thought, and it further deteriorates to 6.2% when the number of inaccurate, noisy thoughts reaches 6. In contrast, with 6 irrelevant noisy thoughts, the accuracy decreases to 25.3%, which is still higher than that of inaccurate thoughts.

Observation F.9 (Model’s robustness to noisy thoughts varies depending on the nature of the task). The NoRa-Symbolic Equal task exhibits a similar trend, where the accuracy decreases as the number of noisy thoughts increases, regardless of the noise type. However, the decline in accuracy is more gradual compared to the NoRa-Math Base-9 task, suggesting that the model’s robustness to noisy thoughts varies depending on the nature of the task.